Woojung Song

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RESEARCH INTERESTS

I study how AI agents behave and how to make them more reliable and better aligned with human needs and values. My work focuses on behavioral evaluation, tool use, and user interaction. I am also interested in social agents that model the values, personalities, and changing behavior of people and fictional characters.

EDUCATION

Seoul National University, Seoul, South Korea — Sep 2025 – Present
Integrated M.S.–Ph.D., Graduate School of Data Science
Advisor: Yohan Jo, HOLI Lab (Human-Oriented Language Intelligence)

Hanyang University, Seoul, South Korea — Mar 2022 – Aug 2025
B.I.S. Department of Information Systems — Overall GPA: 4.35/4.5 | Major GPA: 4.35/4.5

PUBLICATIONS

*Equal contribution. **First** denotes first authorship, including equal contribution.

Source Preference in the Wild: How LLM Agents Favor Items by Source, and How to Reduce It 2026
Under Review
Jonghyun Song*, Haewon Park*, Jeonghoon Shim*, **Woojung Song**, Yohan Jo

AgentHabit: Characterizing Distinct Behaviors of Agents on Everyday Tasks 2026
Under Review · **First**
Woojung Song*, Hoyeol Yang*, Jeonghoon Shim, Sungjib Lim, Jonggeun Lee, Yunho Choi, Yohan Jo

Agents' Overreliance on Unreliable Tools 2026
Under Review · **First**
Hoyeol Yang*, **Woojung Song***, Taewon Kim, Jonghyun Song, Seoyeon Park, Yohan Jo

ArcANE: Do Role-Playing Language Agents Stay in Character at the Right Time? 2026
Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), Main · **First**
Woojung Song*, Nalim Kim*, Sangjun Song, Chaewon Heo, Jongwon Lim, Yohan Jo

Human Psychometric Questionnaires Mischaracterize LLM Behavior 2026
Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), Main · **First**
Woojung Song*, Dongmin Choi*, Yoonah Park, Jongwook Han, Eun-Ju Lee, Yohan Jo

Don't Adapt Small Language Models for Tools; Adapt Tool Schemas to the Models 2026
Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL), Main · **First**
Jonggeun Lee*, **Woojung Song***, Jongwook Han, Haesung Pyun, Yohan Jo

Non-Collaborative User Simulators for Tool Agents 2026
The Fourteenth International Conference on Learning Representations (ICLR)
Jeonghoon Shim, **Woojung Song**, Cheyon Jin, Seungwon Kook, Yohan Jo

Psychometric Item Validation Using Virtual Respondents with Trait-Response Mediators 2026
Transactions of the Association for Computational Linguistics (TACL)
Sungjib Lim, **Woojung Song**, Eun-Ju Lee, Yohan Jo

Quantifying Data Contamination in Psychometric Evaluations of LLMs 2026
Findings of the Association for Computational Linguistics: EACL · **First**
Jongwook Han*, **Woojung Song***, Jonggeun Lee*, Yohan Jo

Value Portrait: Assessing Language Models' Values through Psychometrically and Ecologically Valid Items 2025
Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL), Main · **First**
Jongwook Han*, Dongmin Choi*, **Woojung Song***, Eun-Ju Lee, Yohan Jo

Interpretable Prediction of Private Brand Purchases by Pet Type in E-Commerce for Consumer Behavior Analysis Using Real-World Transaction Data 2026
PeerJ Computer Science · **First**
Jaehyuk Lee*, **Woojung Song***, Jina Kim, Yoona Chung, Eunchan Kim

RESEARCH EXPERIENCE

Seoul National University, HOLI Lab — Seoul, South Korea
Undergraduate Research Intern — Jul 2024 – Aug 2025

- Researched value alignment in Large Language Models (LLMs)
- Developed simulation frameworks using LLMs to model complex human behaviors and interactions

WORK EXPERIENCE

LG Electronics, H&A Sound/Image Intelligence R&D Lab — Seoul, South Korea
Intern — Mar 2024 – Jun 2024

- Enhanced LLM-based Web Question Answering using RAG and multi-agent approaches
- Developed WAAG, an automated QA evaluation system with 40% accuracy improvement
- Achieved highest level in LG SW Competency Certification Test

US 8th Army, Korean Augmentation To the United States Army — Pyeongtaek, South Korea
Sergeant (Senior KATUSA) — May 2020 – Nov 2021

- Served in US 8th Army as part of the mandatory military service program
- Received Certificate of Appreciation for exceptional service

PATENTS

English Problem Creation System Using Multi-Agents — App. No: 10-2024-0109541
Filed: August 16, 2024

- AI system for automated creation of English exam questions
- Aims to generate high-quality content and reduce educator workload

SCHOLARSHIPS

Presidential Science Scholarship for Graduate Students — South Korea
Ministry of Science and ICT — 2026 – 2027

- Selected at a 25:1 competition ratio

AWARDS

Busan Mayor's Award — Busan, South Korea
DIVE Competition, 2nd Place — 2026

- Also won 1st Place from Busan Metropolitan Corporation in the same competition

Chuncheon Mayor's Award — Chuncheon, South Korea
Chuncheon Generative AI Competition, 3rd Place — Jun 2025

- Developed "Blue-Ming" Chuncheon City Youth Entrepreneurship AI Consulting Platform

Busan Mayor's Award — Busan, South Korea
DIVE Competition, App and Service Development, 3rd Place — 2024

- Developed "Korail Donghae" application integrating Language Models with custom tools and RAG
- Also won 1st Place from Korea Railroad Corporation Busan-Gyeongnam Headquarters

SKILLS & COMPETENCIES

- **Programming:** Python, C/C++
- **Language:** Korean (native), English (fluent)

EXTRA-CURRICULAR ACTIVITIES

FORIF (Hanyang Programming Club) — Seoul, South Korea
Member (2022 – 2025), Vice President (2023)

- Organized club events, including HPEC, doubling club membership
- Managed HPEC & HSPC algorithm competition as Project Manager